

Selaine Rodriguez

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EDUCATION

Stanford University

Stanford, CA

Master's of Science in Computer Science; GPA: 3.82

September 2022 - June 2024

Concentration in Human-Computer Interaction

Stanford University

Stanford, CA

Bachelor of Science in Computer Science; Minor: Education; GPA: 3.87

August 2019 - June 2023

Concentration in Human-Computer Interaction

Honors Thesis in Education: Cultural Centers on College Campuses: Understanding Marginalized Students'

Perceptions of Community Centers

Ysleta High School

El Paso, TX

High School Diploma

August 2015 - June 2019

SKILLS

Languages: English, Spanish (Home-Background Speaker)

Coding Languages: Python, C++, C, Golang, JavaScript, Typescript, HTML, CSS

Technologies: Git, React, React Native, MongoDB, NodeJS, Expo, Unity, Blender, Canva

ENGINEERING EXPERIENCE

Uber Technologies Inc.

Bay Area, CA

Software Engineering I

September 2024 - Present

- Web engineer on the Uber Eats for Business (U4B) team
- Maintain codebase and improve engineering quality for group orders, meal plans, meal programs, and other U4B experiences
- Work on expanding business including the 'Saved Groups' feature

Software Engineering Intern

June 2023 - September 2023

- Frontend intern on the Rider Platform with the Rider Web Verticals team
- Created a webpage that improves user experience and integrates new product verticals into the trip management experience
- Used React Native, Typescript, and Base Web UI to create dynamic card elements to showcase different trip elements
- Integrated multiple API backends to access user's past, upcoming, and current trip statuses, which I then cleaned up using Apollo and GraphQL

Software Engineering Intern

June 2022 - September 2022

- Frontend intern on the Rider Platform with the Rider Web Verticals team
- Created a webpage using React Native, JavaScript, and Base Web UI that contains dynamic elements such as an image carousel, product cards, and drop-down navigation bar
- Integrated 10+ Uber products into web page options using a GraphQL Query and Resolver, allowing for different products to be showcased based on user needs

Software Engineering Intern

June 2021 - September 2021

- Backend intern on the Carbon/Driver Platform with the Driver Engineering team
- Worked on a project focused on creating a Server-Driven UI that allows backend developers to compose UI responses and send over to mobile device as a visual model

- Project completion facilitated faster iterations on mobile features by allowing backend engineers to see how the payload will appear on the app without having to push updates to users

TEACHING EXPERIENCE

Lower Valley Scholars Foundation

El Paso, TX

Co-Founder & Director of Outreach

May 2020 - Present

- Established foundation focused on helping first-generation, low-income, minority students from the Lower Valley area of El Paso achieve their educational goals
- Drafted founding documents and assisted with the legal documentation needed to become official non-profit
- Create resources for students in the form of slideshows, workshop presentations, and documents
- Work with local business and entities to develop partnerships and sponsorships
- More information: <https://www.lowervalleyscholarsfoundation.org/>

Stanford Computer Science Department

Stanford, CA

CS 278: Social Computing – Head Teaching Assistant

April 2024 - June 2024

- Managed teaching team of 10 and worked closely with Prof. Michael Bernstein to ensure course flowed smoothly
- Assisted with all aspects of the course including exam creation, website management, course design, grading, student concerns, and more
- Covered topics such as virality, feed ranking, peer production, moderation, misinformation, and other significant social factors of computing
- Class size of 295

CS 107: Computer Organization and Systems – Teaching Assistant

January 2024 - March 2024

- Taught the C language, as well as topics such as Unix, memory allocation, stack/heap design, pointer arithmetic, and bitwise manipulation
- Graded assignments, reviewed code, and ran office hours
- Class size of 255 students

CS 148: Introduction to Computer Graphics – Teaching Assistant

September 2023 - December 2023

- Teach introductory computer graphic concepts such as light, shading, ray-tracing, and more
- Grade assignments and assist students with final projects in Blender
- Class size of 488 students

CS 106A/106B Section Leader

September 2021 - June 2023

- Teach a weekly section, assist with conceptual/debugging office hours, grade student assignments and assessments
- Started DEI initiative of weekly office hours for first-generation and/or low-income students
- Monitored Ed Platform with student questions as Head TA for Stanford Summer Engineering Academy iteration
- Participated in CS Bridge program where I taught introductory topics in Spanish at the Universidad de Los Andes in Bogota, Colombia

Code in Place Section Leader

April 2021 - May 2021

- Acted as a volunteer section leader during a research project/widespread engineering project
- Work with a team of more than 50 teaching leads and 1000 section leaders to support 10,000+ students across the world as they navigate the first five weeks of CS 106A: Programming Methodology
- Prepare materials for and teach a section of 10 students on a weekly basis

Stanford School of Engineering

Stanford, CA

Lecturer

August 2023

- Acted as the main lecturer for the Stanford Summer Engineering Academy's iteration of CS 106A: Programming Methodologies

- Created course assignments that test students on basic coding skills in Python, including for/while loops, variables, booleans, images, grid, conditionals, and more
- Managed the course website as well as monitor the Ed platform with student questions

Stanford Society of Latinx Engineers

Stanford, CA

SOLE LaIR Co-Founder

January 2023

- Established a peer support program for Latinx students in introductory computer science courses
- Provided individualized assistance with debugging and conceptual understanding to boost students' confidence in programming
- Extends traditional LaIR office hours to cater to the needs of underrepresented groups in STEM at Stanford
- Worked as a helper from January 2023 to June 2024

Stanford Center for Teaching and Learning

Stanford, CA

Program Development Associate

January 2021 - September 2022

- Help develop materials for pilot program, IDEAL Pedagogy, which focuses on increasing inclusivity and diversity in higher education
- Create set of asynchronous modules, design website, and develop campus-wide partnerships

Hatch Learning

Bay Area, CA

Computer Science Teacher

August 2020 - January 2020

- Teach computational thinking and computer science to K-2 students weekly using PockeTurtle curriculum

Our Lady of Mt. Carmel Catholic Church

El Paso, TX

Teacher's Assistant

September 2011 - May 2019

- Assisted teacher with developing lesson plans, preparing materials, and teaching 3rd grade students

ADDITIONAL EXPERIENCE

Mariachi Femenil Orgullo Mexicano

Bay Area, CA

Trumpet Player

November 2024 - Present

- Currently on trial with the Bay Area's first all-female mariachi group
- Attend rehearsals once a week and gig on Saturdays
- Repertoire exists of 30+ songs

Stanford Residential Education

Stanford, CA

Residential Assistant (RA)

August 2021 - June 2023

- Create community and promote wellbeing for undergraduate students
- Plan and run on-call events, dorm trips, dorm merchandise, and other initiatives
- 2-time RA with experience in both an all-frosh dorm (West Lagunita) and ethnic-themed dorm (Casa Zapata)

City of El Paso

El Paso, TX

Youth Leader for Census Initiative

March 2019 - June 2019

- Worked with local school districts, business owners, and government officials to organize events to increase participation in 2020 Census
- Created a direct communication line between local government and underserved community of the Lower Valley of El Paso

Legislative Intern

June 2018 - July 2018

- Responded to constituents through phone, email, and personal visits
- Attended and assisted with community and city council meetings
- Worked on pet grooming ordinance in support of animal rights (Ordinance No. 019087)

RESEARCH

Stanford Graduate School of Education

Stanford, CA

Undergraduate Honors Theses

September 2022 - June 2023

- Advised by Dr. Ramon Martinez and Dr. Ari Y. Kelman
- In my thesis, *Cultural Centers on College Campuses: Understanding Marginalized Students' Perceptions of Community Centers*, I explored how marginalized students at Stanford University perceive and engage with cultural centers. Through a mixed-methods approach, including both quantitative Likert-scale and qualitative open-ended questions, I surveyed students about their participation in these centers, their views on the centers' goals, and the impact on their sense of identity and belonging. The results revealed a wide range of experiences: while many students reported a strong sense of community and cultural pride, others called for increased focus on intersectionality, diverse representation, and improved programming. This study bridges the gap in research on the role of cultural centers in fostering inclusion and belonging for marginalized students, particularly within prestigious institutions like Stanford.
- See more, or read the full work [here](#).

Stanford Department of Civil Engineering

Stanford, CA

Hybrid Physical + Digital Spaces for Enhanced Sustainability and Wellbeing

June 2020 - August 2020

- Advised by Dr. Sarah Billington
- Developed surveys to examine connection between materials (artificial vs. natural), lighting (artificial vs. natural), or representations (diverse vs. non-diverse) and sense of belonging, self-efficacy, or pro-environmental attitudes
- Project Summary:  Final Slides - CEE Research 2020

PROJECTS

Foodmarks Mobile App (CS 377U)

May 2024

- Created a mobile application using React, Expo, JavaScript, and Firebase that is designed to help users discover new restaurants, plan social events, vote on dining options, and accommodate dietary restrictions within a group
- Won 'Best Analytics' amongst class groups for our analysis of user's interests
- See full article / findings [here](#)

How Does the Trumpet Work? (EDUC 423)

March 2024

- Used VegaLite, D3, and JavaScript to create an educational explorable that explains the history and physics behind the trumpet, as well as allows the user to play with a trumpet simulator
- Designed all learning exercises and visuals myself, as well as recorded the trumpet audio from my own playing
- See full educational tool [here](#)

Exploring Coffee Production, Consumption, and Characteristics (CS 448B)

December 2023

- Used VegaLite and D3 to create data visualizations that capture the history of coffee, as well as consumption, flavor palettes, and other notable data
- See our full visualization [here](#)

Automatic Crossword Generator (CS 206)

March 2023

- Created a web application using CSS, HTML, and JavaScript that generates various crosswords depending on relevant news categories
- Promotes usage by local newspapers / news sources by allowing them to plug-in their own stories, in hopes to raise reader engagement

Task Together Mobile App (CS 194W)

March 2023

- Created a social media platform focused on encouraging busy students to run more errands together, schedule more social hangouts, and generally spend more time with others
- App consists of social groups you can create, and then invite to different tasks (with in-app support for RSVP, adding / editing event details, tagging events, etc.)
- Created app using JavaScript, Expo, React, and Firebase – with initial designs in Canva
- Managed all HTML, CSS, and other front-end styling

Real-Time 3D Shader (CS 248A)

March 2023

- Implemented lighting and material shading effects using GLSL, OpenGL's shading language and OpenGL client code in C++
- Coding implementations address material pattern logic, material BRDFs and lighting computations to compute surface reflectance

Cardinal3D Graphics Software

February 2023

- Used C++ to extend implementation of Cardinal3D, a 3D graphics software, by adding interactive mesh editing (edge flip, beveling, triangulation, simplification, etc.) and realistic path tracing (ray generation, object intersection, bounding volume hierarchy, shadow rays) functionality

Dia de los Muertos 3D Rendering (CS 148)

December 2022

- Used Blender to create a vibrant, ray-traced digital representation of an ofrenda, incorporating personal and traditional elements such as pan de muerto, photos of loved ones, marigolds, and sugar skulls, to celebrate the holiday's spirit and cultural significance
- Created majority of elements from scratch and used advanced Blender features such as particle systems, volumetrics, UV wrapping, and HDRIs
- See more [here](#)

soRRRt (CS 247G)

May 2022

- Created a PC game in Unity that encourages players to correctly sort their trash into the correct recycling, trash, and compost categories
- Ultimate goal was to promote environmental awareness
- Play [here](#) or learn more about our design process [here](#)

LangVRse (CS 147)

March 2022

- Created a virtual reality game designed to immerse users in a new language and culture, allowing them to practice real-world activities and enhance language fluency through engaging, interactive experiences
- Engaged in the entire design process, including needfinding interviews, experience prototypes, low-fidelity and high-fidelity prototypes, etc.
- Won **Best Concept** and **Third Best Overall Project** (out of 55 teams in course)
- See more [here](#)

Stanford Shell (CS 110)

May 2021

- Created a fully functional Linux shell/terminal in C that allows for basic Linux commands, use of pipelines to transfer data across different commands, the running of programs that have access to the terminal for input, and more

Implicit and Explicit Heap Allocator (CS 107)

March 2021

- Created an implicit heap allocator in C following a first-fit algorithm
- Created an explicit heap allocator in C following a first-fit algorithm and address-based ordered explicit free list
 - Implemented coalescing and in-place reallocation to optimize performance

Top Participants Program (CS 106B)

November 2020

- Created a program in C++ that takes in participant names and a score and returns top participants wanted by user as a final project for CS 106B: Programming Abstractions
- Designed a struct to represent datapoints and a class to represent a priority queue based on a max heap
- See more [here](#)

Bajillion Search Engine (CS 106A)

May 2020

- Created a search engine utilizing lists, dictionaries, classes, local host, and other basic concepts in Python as an extension for a CS 106A: Programming Methodology assignment

Rainbow Sand (CS 106A)

May 2020

- Created an interactive graphics window in Python that allows the user to place rocks and drop sand that changes colors based on its location and movement as an extension for a CS106A: Programming Methodology assignment

MENTORING

Lower Valley Scholars Foundation Mentorship Program

August 2020 - Present

Antonio Gomez	Ysleta HS Student (Class of 2023)
Victoria Ramos	Ysleta HS Student (Class of 2022)
Kianna Leon	Ysleta HS Student (Class of 2021)
Maria Fuentes	Ysleta HS Student (Class of 2021)
Rebecca Candelaria	Ysleta HS Student (Class of 2021)

Society of Latinx Engineers (SOLE) Mentorship Program

November 2020 - June 2024

Efrain Angon-Cruz	Stanford Undergraduate (Class of 2027)
Jolie Barga	Stanford Undergraduate (Class of 2027)
Ginelle Servat	Stanford Undergraduate (Class of 2026)
Guadalupe Soto	Stanford Undergraduate (Class of 2026)
Yamilett Estrada-Reyes	Stanford Undergraduate (Class of 2026)
Kelly Bonillo-Guzman	Stanford Undergraduate (Class of 2025)
Sebastian Ingino	Stanford Undergraduate (Class of 2025)
Yarency Avelar	Stanford Undergraduate (Class of 2025)
Marco Pizarro	Stanford Undergraduate (Class of 2024)

Society of Latinx Engineers (SOLE) Tech Challenge Mentor

January 2020 - May 2024

Sofia	San Jose CA Middle Schooler (3x)
Maya	San Jose CA Middle Schooler (2x)
Andrea	San Jose CA Middle Schooler (2x)
Jenny	San Jose CA Middle Schooler (2x)
Guadalupe	San Jose CA Middle Schooler
Diego	San Jose CA Middle Schooler
Adamari	San Jose CA Middle Schooler
Alejandro	San Jose CA Middle Schooler
Valeria	San Jose CA Middle Schooler
Cruz	San Jose CA Middle Schooler
Jacky	San Jose CA Middle Schooler
Camilla	San Jose CA Middle Schooler
Edgar	San Jose CA Middle Schooler
Jazzlyn	San Jose CA Middle Schooler
Briza	San Jose CA Middle Schooler

Hispanic Scholarship Fund National Leadership Conference		<i>July 2021</i>
Elena Sierra	Stanford Undergraduate	
Ella Serrano-Wu	Harvard Undergraduate	
Jeniffer Maldonado	University of Florida Undergraduate	
Paulina Morales	Bowdoin College Undergraduate	
Victoria Bardon	University of Texas Undergraduate	
The Gates Scholarship Summer Institute		<i>June 2021</i>
Alyssa Matthews	Northwestern Undergraduate	
Elizabeth Kemp	Hamilton Undergraduate	
Grace Chow	Stanford Undergraduate	
Marleni Angel	Brown Undergraduate	
Peyton Brayboy	U.N.C. Chapel Hill Undergraduate	
First-Generation and/or Low-Income Partnership (FLIP) Sibling Program		<i>October 2020 - June 2021</i>
Jenny Mai	Stanford Undergraduate	
Olivia Viruet-Quintero	Stanford Undergraduate	
Senem Isik	Stanford Undergraduate	
The Phoenix Scholars Mentorship Program		<i>June 2020 - June 2021</i>
Lizbeht Pelayo	LAUSD HS Student	
New Student Orientation (NSO) Volunteer Mentor		<i>July 2020</i>
Amaru Ordóñez-Jacobson	Stanford Undergraduate	
Edwin Pua	Stanford Undergraduate	
Jesus Olivares	Stanford Undergraduate	
The Gates Scholarship Summer Institute		<i>June 2020</i>
Brianna Cajamarca	Amherst Undergraduate	
Britney Mayorga	Emory Undergraduate	
Candace Williams	Tufts Undergraduate	
Candy Tang	Stanford Undergraduate	
Elizabeth Kemp	U.T. Austin Undergraduate	
First-Generation and/or Low-Income Partnership (FLIP) FLI Guide		<i>April 2020</i>
Olivia Viruet-Quintero	Stanford Admit / HS Student	
Michelle Fox	Stanford Admit / HS Student	

LEADERSHIP

Stanford First Generation and/or Low-Income Alumni Network (FLAN)		<i>February 2025 - Present</i>
Committee Volunteer Member & Web Developer		
Stanford University		
Mariachi Cardenal de Stanford, Lead Trumpet		<i>September 2020 - June 2024</i>
Society of Latinx Engineers, SOLE LaIR Founder & Assistant		<i>January 2023 - June 2024</i>
Society of Latinx Engineers, President		<i>March 2022 - March 2023</i>
Mariachi Cardenal de Stanford, President		<i>May 2020 - May 2022</i>
Management Leadership for Tomorrow, Career Prep Fellow		<i>March 2021 - December 2022</i>
Society of Latinx Engineers, Vice-President		<i>March 2021 - March 2022</i>
Cap & Gown Women's Leadership, Media Director		<i>June 2020 - June 2021</i>
Society of Latinx Engineers, Social Director		<i>March 2020 - March 2021</i>
Society of Latinx Engineers, Freshman Intern		<i>October 2019 - March 2020</i>

Ysleta High School

National Honor Society, Historian
Pueblo Viejo Neighborhood Association, Honorary Board Member
Superintendent's Advisory Committee
Gay-Straight Alliance, Treasurer
Superintendent's Scholars
Robotics Social Officer
Drum Major
Student Body President
Principal's Advisory Committee
T-STEM Academy, Engineering Ambassador
Class of 2019, Vice President
Technology Student Association, Secretary
Freshman Basketball, Co-Captain

May 2018 - June 2019
November 2017 - June 2019
August 2017 - June 2019
August 2017 - June 2019
August 2017 - December 2017
August 2017 - June 2018
May 2017 - June 2019
May 2017 - June 2019
August 2016 - June 2019
August 2016 - June 2019
August 2016 - June 2017
August 2016 - June 2017
October 2015 - February 2016

HONORS & AWARDS

2024

Lloyd W. Dinkelspiel Award for Distinctive Contributions to Undergraduate Education from Stanford's Annual University / Presidential Awards
Centennial Teaching Award from the Stanford School of Engineering

2023

Outstanding Achievement Award from the Stanford Alumni Association
Award of Excellence from the Stanford Alumni Association
Leader Award from Stanford's Cap and Gown Women's Leadership Organization
SOLE Outstanding Member of the Class of 2023

2022

Jerry I. Porras Award from Stanford's El Centro Chicanx y Latinx
Renato Rosaldo Jr. Award from Stanford's El Centro Chicanx y Latinx
SOLE Outstanding Member of the Year
SOLE Outstanding Member of the Class of 2023

2021

Hispanic Scholarship Fund Scholar of the Year Finalist
SOLE Outstanding Member of the Year
SOLE Outstanding Member of the Class of 2023
SOLE Leadership in Outreach

2020

CODE 2040 Fellow Finalist
SOLE Outstanding Underclassmen of the Year
SOLE Outstanding Member of the Class of 2023
Highest Academics Honors from Stanford's El Centro Chicanx y Latinx
Outstanding Psychology 1 Scholar

2019

Gates Scholar
Hispanic Scholarship Fund Scholar
National Hispanic Scholar
Burger King McLamore Scholar
AP Scholar with Distinction

El Paso Borderplex STEAMX Scholar
Ysleta Education Foundation Distinguished Scholar
FirstLight Federal Credit Union Scholar
Altar Server of the Year at Our Lady of Mt. Carmel
Texas State Science Fair Qualifier
Valedictorian

PROFESSIONAL EVENTS, TALKS & AFFILIATIONS

Professional Events

Society of Hispanic Professional Engineers National Conference	<i>2020, 2022, 2023</i>
HackMIT	<i>September 2020</i>
ACM Richard Tapia Celebration of Diversity in Computing Conference	<i>September 2020</i>
Hispanic Scholarship Fund National Leadership Conference	<i>July 2020</i>
UTEP NEXUS Engineering Program	<i>July 2017</i>
El Paso Water Symposium	<i>June 2017</i>

Professional Affiliations

Hispanic Scholarship Fund
Society of Hispanic Professional Engineers (SHPE)
The Gates Scholarship

Talks

Stanford Lambda Theta Nu's Latina Youth Leadership Conference	<i>April 2022, April 2023</i>
How to Start Your Own Non-Profit	